Dataful's Cleaning and Reconciliation Strategy

Central Public Sector Enterprises (CPSEs) are central to research on public finance, employment, investment and disinvestment, which increases the need for granular and reliable CPSE data. The Department of Public Enterprises (DPE) and stock-market sources publish extensive data on CPSEs, but the same enterprise is often written in many different ways across tables, years and publications. At Dataful, we have put in significant effort to clean up, reconcile and standardise this data so that every CPSE can be tracked as one entity over time.

Managing CPSE data accurately is challenging because enterprises change over time. CPSEs are renamed, merged, demerged, listed or delisted, set up subsidiaries and joint ventures, and sometimes move out of government ownership through disinvestment. If these changes are not accounted for, they result in duplicate records and an inaccurate picture of the number and performance of unique enterprises.

Since there is no single authoritative list that links every CPSE name, subsidiary and legal identifier, Dataful follows a systematic approach to identify and eliminate such duplications. By tracking events such as name changes, mergers, demergers, changes in listing status and changes in ownership, Dataful reconciles historical records to create a consistent and reliable view of CPSEs across time.

The sections below explain the methodology. Before that, it is important to understand how a company's Corporate Identification Number (CIN) is built and when it changes, as the CIN forms the basis for identifying and tracking legal entities over time.

What Does a Company's Corporate Identification Number (CIN) Tell Us?

A **CIN** is a unique 21-character alphanumeric identifier assigned by the Ministry of Corporate Affairs (MCA) at the time of a company's incorporation. It encodes the listing status, industry classification (NIC), state of registration, year of incorporation, ownership type and registration number. The table below decodes the CIN of Bharat Heavy Electricals Limited.

Table 1: Structure of a CIN (example: L74899DL1964GOI004281)

Part	Component	Example	Meaning
1	Listing status	L	L = listed, U = unlisted
2–6	Industry (NIC) code	74899	5-digit National Industrial Classification code
7–8	State code	DL	State or UT of the registered office (Delhi)
9–12	Year of incorporation	1964	Year in which the company was incorporated
13–15	Ownership type	GOI	Type of company at incorporation (see Table 2)
16–21	Registration number	004281	6-digit number issued by the Registrar of Companies

The three-letter ownership code is particularly relevant for CPSEs. The codes that appear in Dataful's CPSE master are listed below.

Table 2: Ownership codes in the CPSE master

Code	Meaning	Typical CPSE-related entities
GOI	Government of India company	CPSEs, their subsidiaries and many project SPVs
SGC	State Government company	State-owned corporations and state-promoted banks

Code	Meaning	Typical CPSE-related entities
PLC	Public limited company (non-government)	Listed and unlisted public companies, including some PSU subsidiaries and disinvested companies
PTC	Private limited company	Joint ventures and subsidiaries incorporated as private companies
NPL	Not-for-profit (Section 8) company	Government-promoted trade promotion and development bodies
FLC	Foreign company	Foreign-registered entities with an Indian presence

When Does a CIN Change?

While a CIN generally remains unchanged throughout the life of a company, the following statutory events result in a modified CIN:

- **Change in listing status:** when an unlisted company becomes listed, or a listed company is delisted, the 1st character changes from U to L or vice versa.
- **Change in registered office from one state to another:** the state code changes (Section 13 of the Companies Act, 2013).
- **Conversion between private and public company:** the company type component of the CIN is updated (Section 14 of the Companies Act, 2013).
- **Change in principal business activity:** where the MCA updates the NIC code, the CIN is updated to reflect it.

Changes that do not result in a new CIN: change in company name, change in directors or key managerial personnel, change in authorised or paid-up share capital, relocation of the registered office within the same state, and **change in shareholding pattern**.

Important note for CPSEs: because a change in shareholding does not change the CIN, the ownership code can lag behind actual ownership. For example, Air India Limited still carries the GOI code even though it moved to private ownership in January 2022. Dataful therefore does not rely on the ownership code alone to classify an enterprise as government or private (see Ownership Classification below).

Dataful's Data Cleaning Strategy

Dataful addresses duplication and inconsistency in CPSE data through several methods:

- **Standardization:** Dataful cleans unstructured or inconsistently formatted names so that one enterprise does not appear as several entities. Abbreviations are expanded and formatting is unified using fixed conventions: Ltd. to Limited, Pvt. to Private, Corpn. to Corporation, Co. to Company, "&" and "And" to and, and Of to of.

Table 3: Examples of name standardization

Name as per source	Standard name
AAI CARGO LOGISTICS & ALLIED SERVICES COMPANY LTD.	AAI Cargo Logistics and Allied Services Company Limited
AAI Cargo Logistics and Allied Services Company Limited	AAI Cargo Logistics and Allied Services Company Limited
Andrew Yule & Co.Ltd.	Andrew Yule and Company Limited
ANDREW YULE & COMPANY LTD.	Andrew Yule and Company Limited

- **Entity Normalisation:** Variations of the same company name (for example, full versus abbreviated or acronym versions) are grouped together as a single entity. The standard name is the full, expanded name of the same company, not its newer or shorter brand name. For instance, "REC Limited" is mapped to "Rural Electrification Corporation Limited". When a new row is added to the source, only blank standard names are filled and

existing standard names are left untouched, so that earlier work stays stable.

- **CIN Mapping:** Every standard name is mapped to its CIN. Where the same company has more than one CIN, typically because of a change in listing status, the additional CIN is recorded separately so that both identifiers resolve to one entity.
- **Market Listing Reconciliation:** The system identifies and merges records when a company's listing status changes (the L and U versions of the same CIN) to ensure continuity.
- **Subsidiaries, Joint Ventures and SPVs:** Subsidiaries, joint ventures and special purpose vehicles have their own CIN and are retained as separate entities, linked to the parent through the standard name. Several CPSE subsidiaries are project SPVs, for example transmission companies set up by PFC Consulting and REC Power Development and Consultancy, which are later transferred to the winning bidder.
- **Non-company records:** Source tables sometimes contain rows that are not enterprises, such as exchange-traded fund and receipt heads. Such rows are flagged instead of being treated as CPSEs, so that counts of enterprises are not inflated.
- **Merger Management:**

For merged companies: Dataful retains them as a single entity, i.e., after the merger, all historical and future data is associated with the successor company name. For example, Allahabad Bank merged with Indian Bank on 1 April 2020. However, the official sources may report data separately for both the merged entity and the successor, which can lead to duplication if they are not treated as a single entity throughout the entire time series.

For demerged companies: Dataful retains them as separate entities, i.e., after the demerger, all future data is mentioned separately for both from the year of demerger.

- **Status Attributes:** Maharatna, Navratna and Miniratna status, as designated by the DPE, is maintained as an attribute of the standard entity.